

Lecture 3: Allocation Algorithms

Fundamentals of Operating System Design (CEUC301)

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Learning Outcomes

- Explain memory allocation algorithms.
- Apply First Fit algorithm.
- Evaluate allocation efficiency.

Why Allocation Algorithms?

- Select a suitable free memory block.
- Improve memory utilization.
- Reduce fragmentation.
- Improve system performance.

First Fit Algorithm

- Scan free blocks from the beginning.
- Allocate the first block large enough.
- Simple and fast.
- May leave small holes near the beginning.

First Fit Example

Free Blocks: 100 KB, 500 KB, 200 KB, 300 KB, 600 KB

Process = 212 KB

- 100 KB → Not sufficient
- 500 KB → Allocated
- Remaining free block = 288 KB

Best Fit Algorithm

- Searches all free blocks.
- Allocates the **smallest** block that can satisfy the request.
- Reduces leftover space for the current allocation.
- May create many small unusable holes.

Best Fit Example

Free Blocks:

100, 500, 200, 300, 600 KB

Process Size = 212 KB

- 300 KB is selected.
- Remaining free block = 88 KB.

Worst Fit Algorithm

- Allocates the **largest** available free block.
- Leaves a relatively large remaining hole.
- Attempts to avoid many tiny fragments.
- Often performs worse than First Fit in practice.

Worst Fit Example

Free Blocks:

100, 500, 200, 300, 600 KB

Process Size = 212 KB

- 600 KB is selected.
- Remaining free block = 388 KB.

Next Fit Algorithm

- Similar to First Fit.
- Search starts from the location of the previous allocation.
- Reduces repeated scanning from the beginning.
- Performance is usually close to First Fit.

Next Fit Example

Free Blocks:

100, 500, 200, 300, 600 KB

Assume previous allocation ended at the 200 KB block.

Process Size = 250 KB

- Search starts after 200 KB.
- 300 KB block is selected.
- Remaining free block = 50 KB.

Comparison of Allocation Algorithms

Algorithm	Speed	Memory Use	Remarks
First Fit	High	Good	Most widely used
Best Fit	Low	Moderate	Creates many small holes
Worst Fit	Low	Moderate	Leaves large holes
Next Fit	High	Good	Faster search continuation

Numerical Problem

Free Blocks:

100, 500, 200, 300, 600

Processes:

212, 417, 112, 426

Find the allocation using:

- First Fit
- Best Fit
- Worst Fit
- Next Fit

Algorithm Selection Guidelines

Requirement	Recommended Algorithm
Fast allocation	First Fit / Next Fit
Better space utilization	Best Fit
Large remaining holes	Worst Fit
General-purpose OS	First Fit

Summary

- Allocation algorithms select suitable free memory blocks.
- First Fit is simple and widely used.
- Best Fit minimizes immediate waste.
- Worst Fit leaves the largest remaining block.
- Next Fit continues searching from the previous allocation.

Questions

- 1 Explain First Fit with an example.
- 2 Compare First Fit and Best Fit.
- 3 Differentiate Best Fit and Worst Fit.
- 4 Explain Next Fit.
- 5 Solve a numerical problem using all four algorithms.